Faculty of Computing

INTERNSHIP REPORT

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1 INTRODUCTION

1.1 Company Overview

HashBaze is a completely remote technology firm that offers innovative digital solutions to both local and international clients. HashBaze has clients from Sri Lanka, the United Arab Emirates (UAE), Poland, Australia, Canada, and Denmark, thus indicating a very effective presence of the firm globally.

HashBaze operates in a variety of fields, including web application development, SaaS solutions, AI-based solutions, cloud-based solutions, and research-based digital solutions. HashBaze is committed to the design of highly efficient and effective systems that meet the different needs of their clients.

HashBaze has a collaborative and agile work culture, where the engineering, designing, and product teams work together in a completely distributed environment. There is a focus on innovation, quality assurance, and problem-solving while maintaining a professional engineering culture and software development processes.

HashBaze has different teams, including the Product Engineering Team, Backend Team, Frontend Team, and AI/ML Team. HashBaze's completely distributed and flexible work environment provided me with a very valuable hands-on experience of the real world of software development and a better understanding of the professional engineering culture of a globally distributed team.

1.2 Project Overview

During my internship at HashBaze, I gained hands-on experience working on various projects across different domains, including transportation technology, research management systems, and analytical web applications.

One of the projects that I contributed to during my internship at HashBaze includes a shared taxi service web platform known as Carvee. The project's objective is to enhance and improve the current state of urban transportation by allowing users to book and share rides efficiently. The project includes various functionalities such as ride coordination, user management, and booking.

Another project that I contributed to includes AIMS Innovations, which provides a research management system that allows students, researchers, and institutions to efficiently manage research activities and projects. The project includes various functionalities such as data management, handling surveys, analytics, AI, and storage. I assisted in improving various project features, user interfaces, and testing to ensure smooth functionality and operation.

My main contribution has been to the Gen Me Out web application, which is a board strategy explanation web app used to analyze board positions and explain strategies. I have developed several user interfaces and system functionalities to assist the board analysis and explanation processes. I have also added Large Language Model (LLM) capabilities to the system to enrich intelligent strategy explanations and user interactions. Furthermore, I have worked on UI enhancements to optimize the user experience and system responsiveness. I have also worked on the deployment of the system, including the generation of Dockerfiles and deploying the application to Google Cloud Run for execution. This project is ongoing, and Phase 1 has been completed successfully.

1.3 Glossary

Following are the abbreviations and definitions used in this document.

Abbreviation	Description
FE	Frontend
BE	Backend
API	Application Programming Interface
UI/UX	User Interface / User Experience
DB	Database
SLDC	Software Development Life Cycle

2 INTERNSHIP INSIGHT

2.1 Objectives

This internship experience at HashBaze has been carefully designed to offer comprehensive exposure to the best practices of software engineering while making significant contributions to the ongoing software development projects.

Primary Objective

To bridge the gap between the theoretical knowledge acquired during study and the practical application of the same in the software industry.

Technical Objectives

This includes the following objectives:

- Practical experience in full-stack software development.
- Understanding the system architecture and API integration.
- Understanding the workflows of feature implementation.
- Development of debugging skills.
- Understanding containerization and cloud deployment using Docker and Google Cloud Run.
- Understanding the implementation of Large Language Models (LLMs) in web applications.

Professional Development Objectives

This includes the following objectives:

- Development of communication and collaboration skills.
- Development of effective time management and task prioritization skills.
- Understanding the version control and software development workflows.
- Understanding the professional coding standards.

Objectives of the Internship from the Company's Point of View

This includes the following objectives:

- Support the ongoing software development projects such as Carvee, AIMS Innovations, and Gen Me Out.
- Support the feature implementation, UI enhancement, and debugging of the software.
- Continuously keep the software development momentum going.
- Assess the technical ability, hard work, and flexibility of the intern.

2.2 Procedures

The HashBaze internship was conducted entirely online, and there was a distinct set of development practices that ensured seamless collaboration, consistent progress, and adherence to professional engineering standards.

Onboarding

The onboarding process provided me with access to the necessary tools and resources required to begin working on the project. The initial onboarding process gave me time to familiarize myself with the project's architecture, the layout of the code, and the development workflow. The onboarding process was challenging as it required me to understand the design patterns used in the system, the technology stack used by the organization, and the coding standards that are adhered to.

Communication Framework

A structured communication framework was maintained throughout the internship to ensure transparency and collaboration:

- **Slack:** Used as the primary communication platform for daily updates, technical discussions, task clarifications, and team coordination.
- **Google Meet:** Used for online meetings, requirement discussions, progress reviews, and technical walkthrough sessions.

This framework ensured continuous feedback, real-time clarification of doubts, and smooth coordination across distributed team members.

Development Workflow

The development process followed structured and iterative practices aligned with Agile principles.

- **Daily Stand-Ups:** Conducted via Google Meet to discuss progress, planned tasks, and blockers.
- **Requirement Clarification:** Tasks were discussed and confirmed before implementation.
- **Incremental Development:** Features were developed in stages and tested locally.

- **Version Control:** Git and GitHub were used with proper branching and commit practices.
- **Code Reviews:** Pull requests were reviewed to maintain code quality.
- **Bug Fixing:** Issues were resolved and refined based on feedback and testing.

2.3 Methodology

During my internship at HashBaze, tasks were executed using structured development methods, collaborative tools to ensure efficient workflow and high-quality deliverables.

Development Methodology

- **Agile-Inspired Workflow:** An iterative development approach was followed, where tasks were implemented in short cycles with continuous feedback. Daily stand-ups ensured progress tracking and alignment.

- **Requirement Analysis:** Each task began with requirement clarification and understanding of expected functionality before implementation.

- **Incremental Implementation:** Features were developed in stages, tested locally, and refined before integration into the main branch.

- **Testing & Debugging:** Functional validation and bug fixing were conducted before submitting code for review.

Tools & Technologies

- **Frontend Development:** Implemented multiple UI components and enhanced user interfaces for better usability and responsiveness.

- **Backend Integration:** Integrated APIs and developed core functionalities supporting application logic.

- **LLM Integration:** Incorporated Large Language Model capabilities within the Gen Me Out project to generate intelligent board strategy explanations.

- **Version Control:** Git and GitHub were used for source code management, branching strategies, and pull request workflows.

Collaboration

- **Communication:** Slack and Google Meet were used for task coordination, daily stand-ups, and technical discussions.

- **Code Review:** Peer review processes were followed to maintain code quality and adherence to standards.

- **Continuous Improvement:** Feedback was incorporated to refine features and enhance overall system performance.

3 LEARNING OUTCOME

3.1 Learning

During my internship at HashBaze, I gained comprehensive hands-on experience in both frontend and backend development, along with cloud deployment and AI integration.

Technical Learning

- Implemented multiple frontend UI components and enhanced user interaction workflows.
- Developed backend functionalities including subscription plan management logic.
- Integrated Stripe payment gateway to handle secure online payments and subscription processing.
- Worked with API integration between frontend and backend systems.
- Integrated Large Language Model (LLM) capabilities within the Gen Me Out project to generate intelligent strategy explanations.
- Gained practical experience in Docker-based containerization.
- Deployed containerized applications to Google Cloud Run.
- Improved debugging, validation, and system refinement techniques in production-level applications.

Professional and Soft Skill Development

- Strengthening communication skills through daily stand-ups and technical discussions.
- Improved collaboration within a fully remote engineering team.
- Enhanced time management and task prioritization abilities.
- Developed confidence in handling real client-facing systems and payment integrations.

3.2 Measurable Outcome

Area of Work	Tasks Completed	Outcome
Frontend Development	Implemented multiple UI modules	Improved usability and interaction
Backend Development	Developed plan management logic	Enabled subscription-based workflows
Payment Integration	Integrated Stripe gateway	Enabled secure payment processing
Gen Me Out	UI implementation, core functions, LLM integration	Phase 1 successfully completed
Deployment	Docker & Cloud Run deployment	Scalable cloud-based execution

3.3 Effectiveness of the Effort

The internship at HashBaze was highly effective in bridging academic knowledge with real-world software engineering practices. It provided practical exposure to full-stack development, backend payment integration, cloud deployment, and AI-based system enhancement. Working on live production systems such as Carvee, AIMS Innovations, and Gen Me Out allowed me to understand how theoretical concepts are implemented in professional environments.

Through active involvement in UI development, backend plan management logic, Stripe payment gateway integration, LLM-based functionality, and containerized deployment, I gained hands-on experience in managing end-to-end feature development cycles.

Application of Academic Knowledge

Several academic concepts were directly applied during the internship:

- **Software Development Life Cycle (SDLC):** Applied structured development phases including requirement analysis, implementation, testing, and deployment.
- **Object-Oriented Programming (OOP):** Used modular design and structured coding practices in both frontend and backend development.
- **Database and System Design:** Applied knowledge of data flow and system integration while implementing subscription plan management and payment workflows.
- **Cloud Computing Concepts:** Applied theoretical knowledge of containerization and scalable deployment through Docker and Google Cloud Run.
- **Reflection and Areas for Improvement**

If given the same opportunity again, I would focus more on early architectural planning and performance optimization, particularly for backend services and payment workflows. I would also improve documentation practices and participate more actively in system-level design discussions to strengthen my architectural understanding.

Inclusivity and Engineering Practice

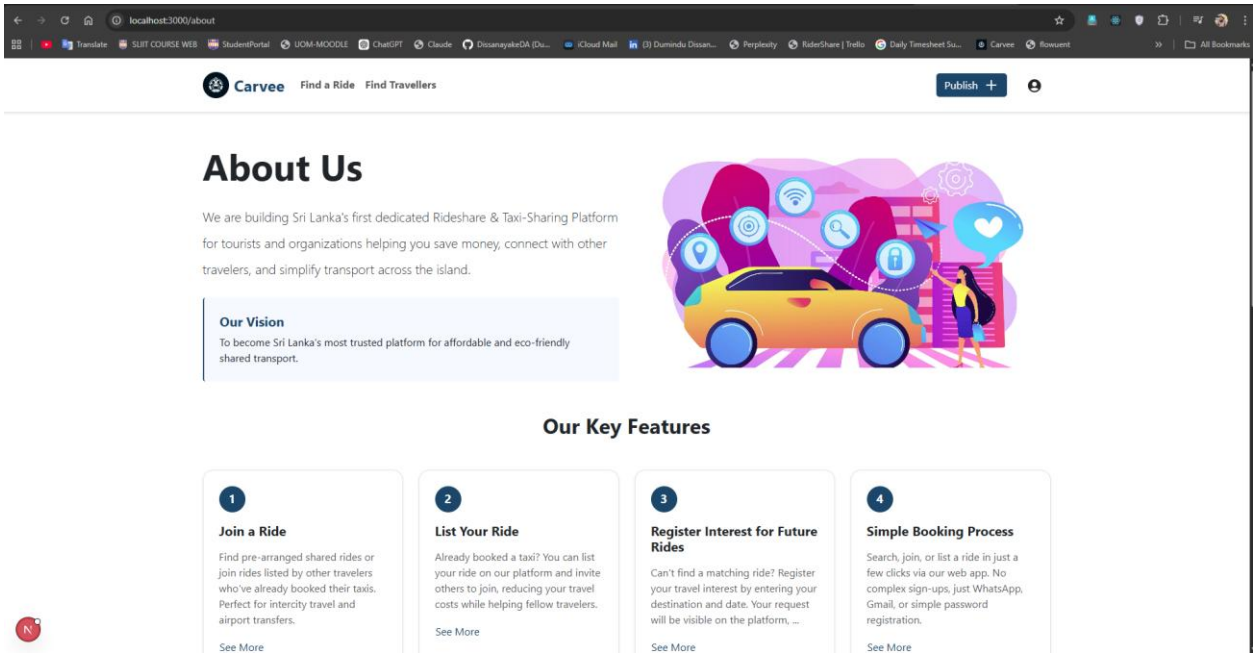
HashBaze maintains a collaborative and inclusive remote work culture. Team members are encouraged to contribute ideas regardless of experience level. Regular communication through Slack and Google Meet, along with peer review processes, ensured equal participation and continuous learning. The supportive engineering environment significantly contributed to my professional growth.

Overall Impact

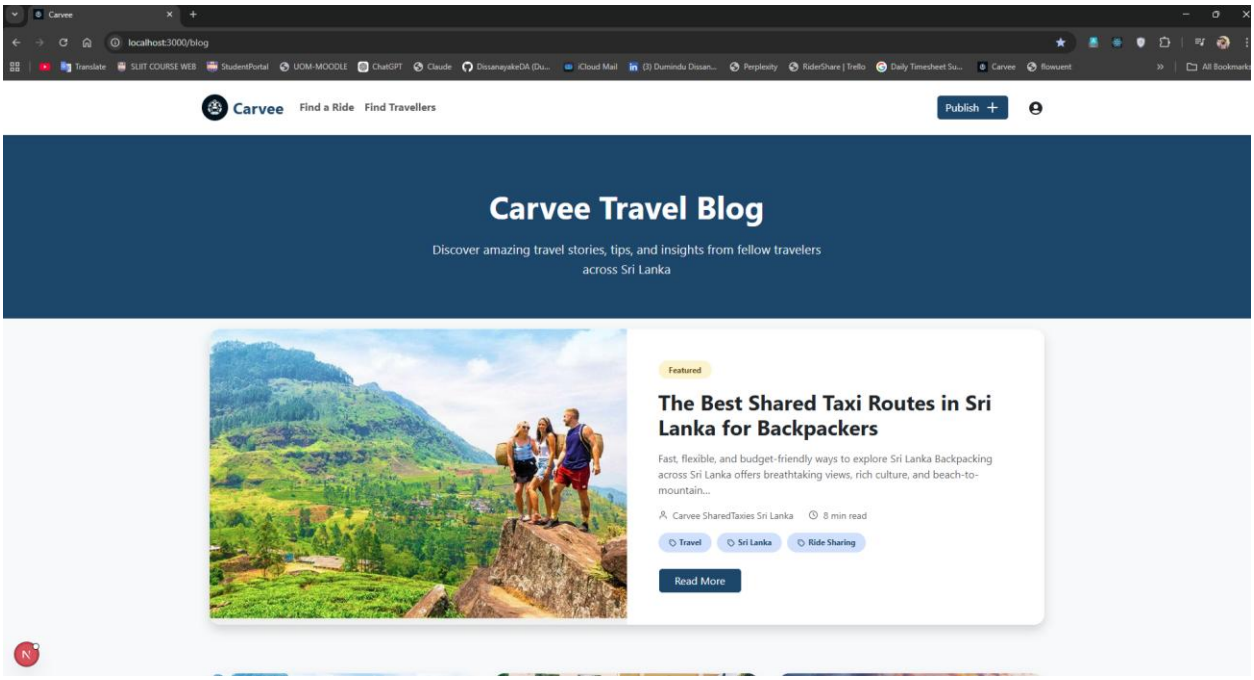
Overall, the internship strengthened my technical expertise in full-stack development, payment gateway integration, LLM integration, and cloud deployment. It enhanced both my professional confidence and practical engineering capability, preparing me for future responsibilities in software development roles.

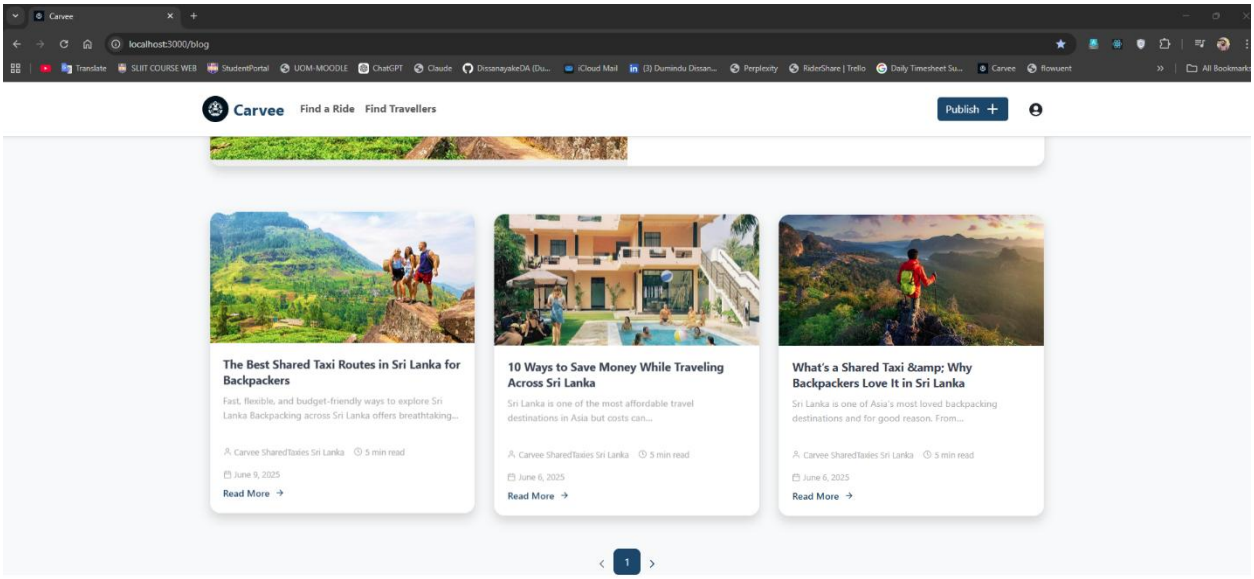
4 SAMPLE WORK

4.1 Carvee About us page revamp



4.2 Carvee Blog Page




Carvee
localhost:3000/blog
Carvee
Find a Ride
Find Travellers
Publish +
The Best Shared Taxi Routes in Sri Lanka for Backpackers
Fast, flexible, and budget-friendly ways to explore Sri Lanka Backpacking across Sri Lanka offers breathtaking...
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Sri Lanka is one of the most affordable travel destinations in Asia but costs can...
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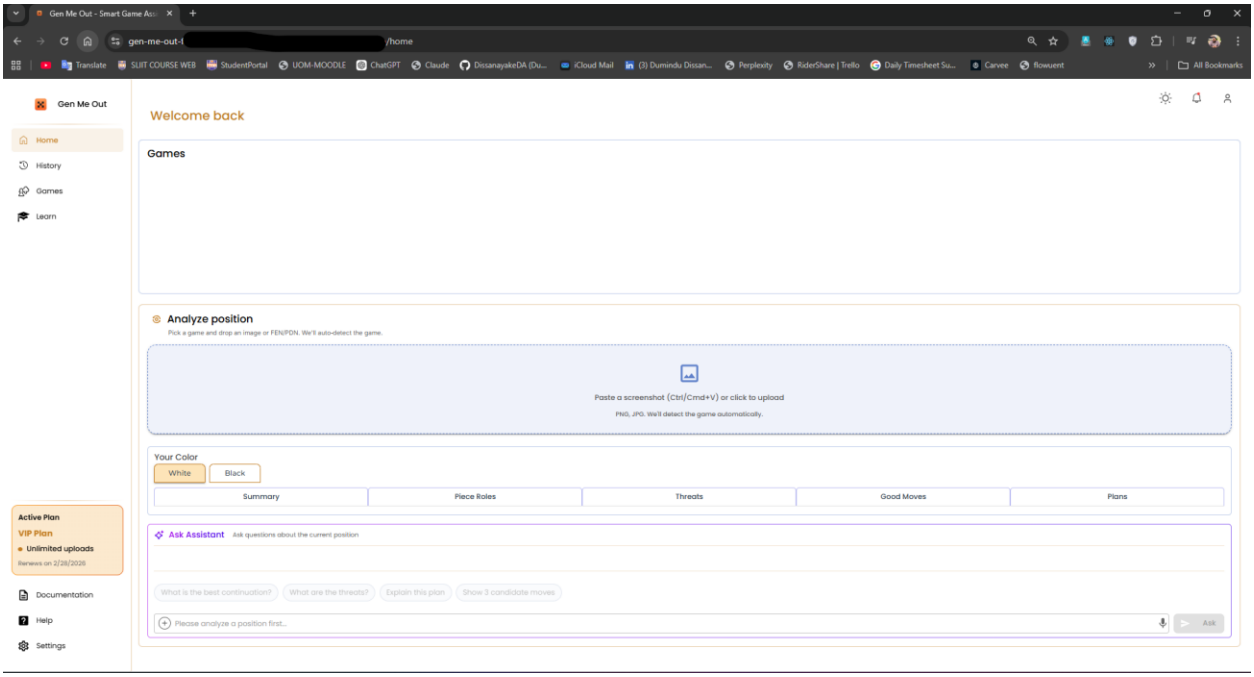
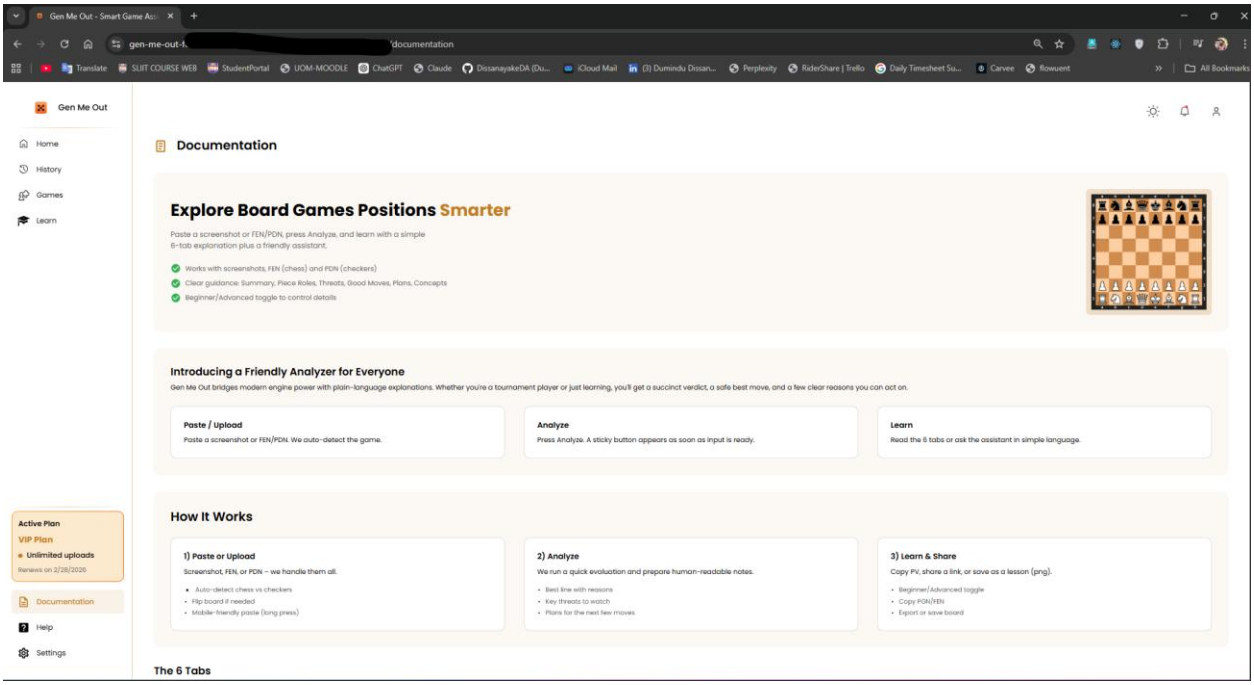
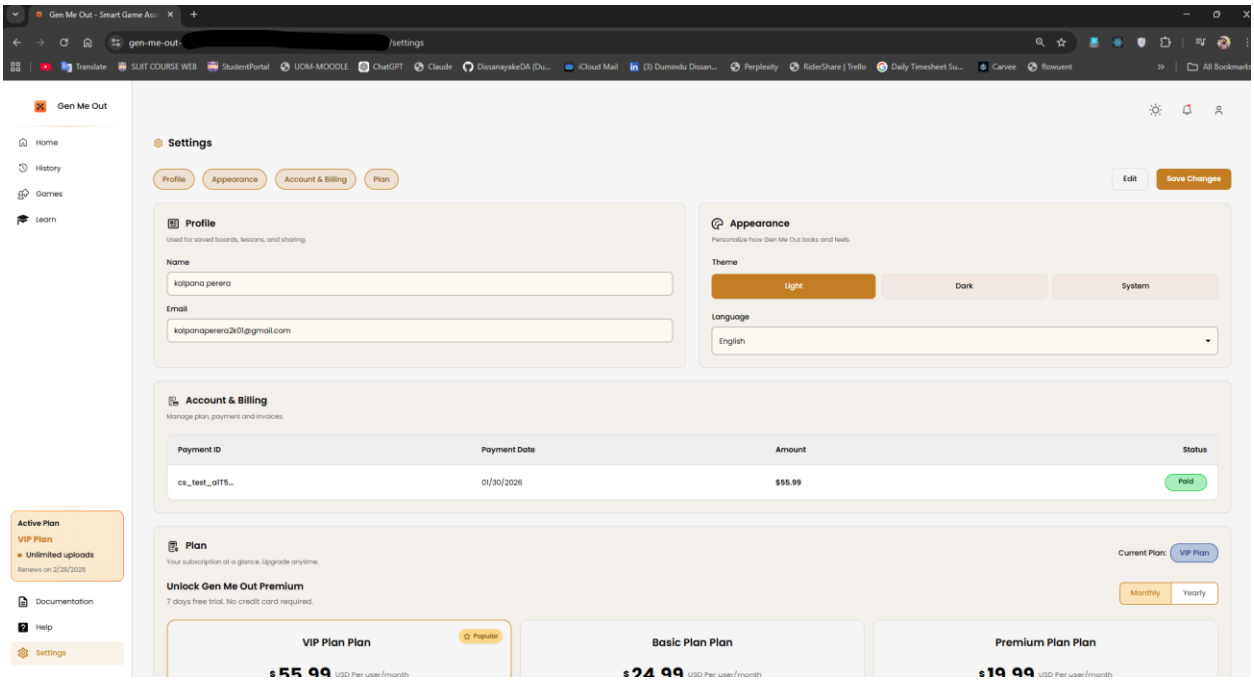
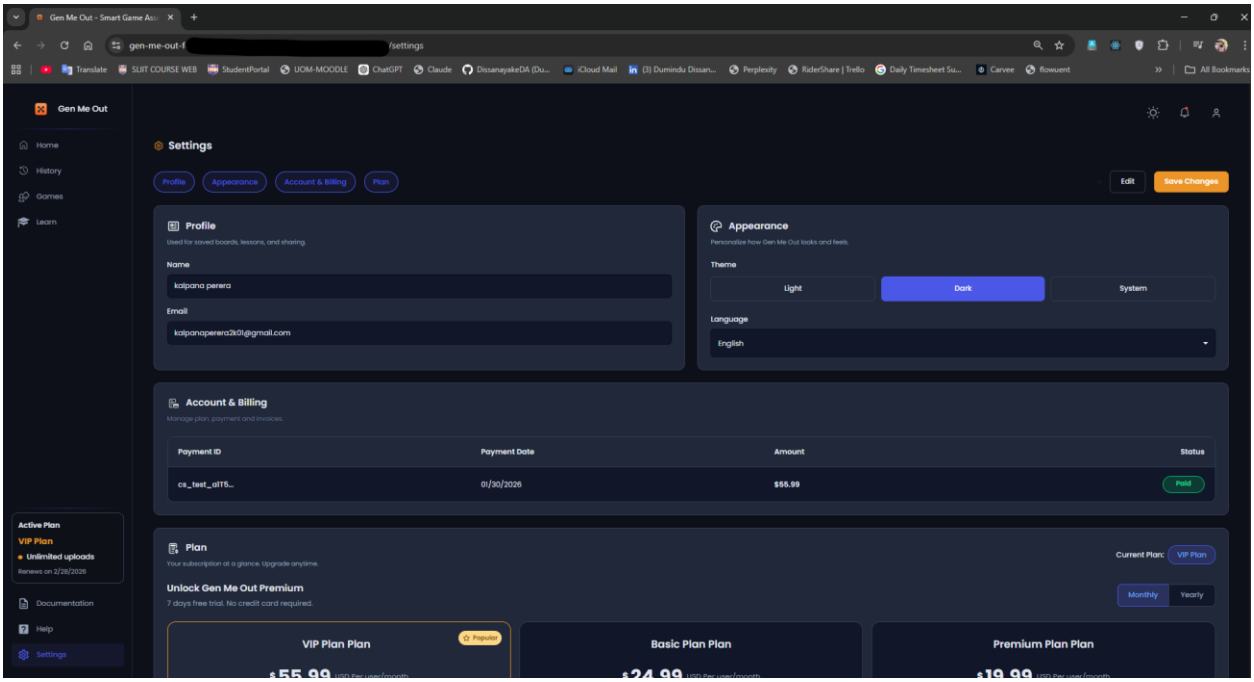
4.4 Gen Me Out


Gen Me Out - Smart Game Ass
/home
Gen Me Out
Home
History
Games
Learn
Welcome back
Games
Analyze position
Pick a game and drop an image or FEN/PDN. We'll auto-detect the game.
Paste a screenshot (Ctrl/Cmd+V) or click to upload
PNG, JPG. We'll detect the game automatically.
Your Color
White
Black
Summary
Piece Roles
Threats
Good Moves
Plans
Active Plan
VIP Plan
Unlimited uploads
Renews on 2/28/2026
Ask Assistant
Ask questions about the current position
What is the best continuation?
What are the threats?
Explain this plan
Show 3 candidate moves
Please analyze a position first...
Ask
Documentation
Help
Settings


Gen Me Out - Smart Game As
gen-me-out-[redacted]/documentation
Gen Me Out
Home
History
Games
Learn
Documentation
Explore Board Games Positions Smarter
Paste a screenshot or FEN/PDN, press Analyze, and learn with a simple 6-tab explanation plus a friendly assistant.
Works with screenshots, FEN (chess) and PDN (checkers)
Clear guidance: Summary, Piece Roles, Threats, Good Moves, Plans, Concepts
Beginner/Advanced toggle to control details
Introducing a Friendly Analyzer for Everyone
Gen Me Out bridges modern engine power with plain-language explanations. Whether you're a tournament player or just learning, you'll get a succinct verdict, a safe best move, and a few clear reasons you can act on.
Paste / Upload
Paste a screenshot or FEN/PDN. We auto-detect the game.
Analyze
Press Analyze. A sticky button appears as soon as input is ready.
Learn
Read the 6 tabs or ask the assistant in simple language.
How It Works
1) Paste or Upload
Screenshot, FEN, or PDN – we handle them all.
Auto-detect chess vs checkers
Flip board if needed
Mobile-friendly paste (long press)
2) Analyze
We run a quick evaluation and prepare human-readable notes.
Best line with reasons
Key threats to watch
Plans for the next few moves
3) Learn & Share
Copy PV, share a link, or save as a lesson (png).
Beginner/Advanced toggle
Copy PGN/FEN
Export or save board
Active Plan
VIP Plan
Unlimited uploads
Renews on 2/28/2026
Documentation
Help
Settings
The 6 Tabs


Gen Me Out - Smart Game As
gen-me-out-[redacted]/settings
Gen Me Out
Home
History
Games
Learn
Settings
Profile
Appearance
Account & Billing
Plan
Edit
Save Changes
Profile
Used for saved boards, lessons, and sharing.
Name
kalpana perera
Email
kalpanaperera2k01@gmail.com
Appearance
Personalize how Gen Me Out looks and feels.
Theme
Light
Dark
System
Language
English
Account & Billing
Manage plan, payment and invoices.
Payment ID
Payment Date
Amount
Status
cs_test_a1T5...
01/30/2026
$55.99
Paid
Plan
Your subscription at a glance. Upgrade anytime.
Current Plan: VIP Plan
Unlock Gen Me Out Premium
7 days free trial. No credit card required.
Monthly
Yearly
VIP Plan Plan
Popular
$55.99 USD Per user/month
Basic Plan Plan
$24.99 USD Per user/month
Premium Plan Plan
$19.99 USD Per user/month
Active Plan
VIP Plan
Unlimited uploads
Renews on 2/28/2026
Documentation
Help
Settings



- Some of the components of the project have been included in the report to show the contributions I made during the internship. Some of the information has been withheld due to the confidentiality agreement of the HashBaze clients. Only the general workflows have been included in the report.